



ashwa MOBILITY
FOUNDATION

LIVE IT. LOVE IT. RACE IT.

Driven by
Legacy,
Remembered
Forever.



**MARCH
2025
EDITION**



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**VISION
AND
MISSION**

GUIDING PURPOSE,
INSPIRING FUTURE.

MISSION AND VISION

VISION

- To be the national leader in all categories of Formula Student, to push the limits of engineering and compete in events at the highest level.
- To foster student mobility solutions in the future by developing sustainable prototypes across various domains.

MISSION

- Prepare students for the industry by designing and building world-class vehicles while facilitating industry interaction.
- Showcase India's engineering expertise in mobility on a global stage and provide talent opportunities for the industry.
- Develop cost-effective, sustainable transportation solutions for the masses.



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**SIDDHARTH
MANDAL
MEMORIAL**

HONORING THE LEGACY, INSPIRING THE FUTURE —
REMEMBERING ASHWA'S FINEST ENGINEERS.



HONORING SIDDHARTH MANDAL: A MEMORIAL TRIBUTE

On February 22nd, the Ashwa community came together at the MCA Seminar Hall to honor and remember Siddharth Mandal, one of Ashwa's founding members and a key figure in its steering committee. His passing earlier this year was a profound loss, and this memorial served as a heartfelt tribute to his enduring impact.



Siddharth was not just a visionary leader but also a dedicated mentor and an inspiration to many. His contributions to Ashwa helped shape its foundation, and his guidance played a crucial role in steering the organization towards success.

The memorial event saw a gathering of Ashwa members, alumni, and well-wishers, all united in remembering Siddharth's legacy. The atmosphere was filled with warmth, respect, and deep emotions as colleagues and friends shared their experiences and spoke about his invaluable influence.

Through moving speeches, shared memories, and moments of reflection, the event became more than just a farewell—it was a celebration of a life well-lived, a legacy that will continue to inspire future generations of Ashwa members.

Though Siddharth is no longer with us, his vision, dedication, and spirit will forever remain a part of Ashwa.





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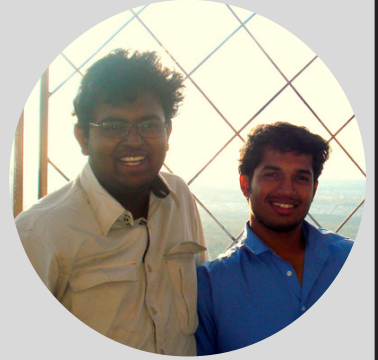
**REMEMBERING
SIDDHARTH
MANDAL**

HONORING A FRIEND, TREASURING THE
MEMORIES



Marvin

2009 BATCH- ENGINE



“The Legacy of Siddharth Mandal: A Lesson in Passion and Innovation

Most of us remember Mandal for his brilliance in finance and his key role in securing Business Awards for Team Ashwa. But today, I want to highlight something even more important—his technical genius and relentless determination.

When I saw the current 2025 car, the first thing that caught my eye was the air intake. It looked strikingly familiar—almost as if Mandal himself had designed it. And in many ways, he had.

Mandal and I worked together on the Engine team, where he was given the challenge of designing the air intake. Before 2006, our intake system was a complete disaster—it looked like an octopus with four carburetors, and the engine’s performance was abysmal. While others accepted this as the status quo, Mandal refused to settle. This was personal for him.

With the arrival of a new fuel-injected Honda engine, Mandal saw an opportunity. He burned the midnight oil, diving deep into Computational Fluid Dynamics (CFD) long before it became common practice. His Venturi design produced incredible results, setting a new benchmark for performance. He went further—sourcing a high-grade 7000-series aluminum alloy and overseeing every step of the manufacturing process to ensure perfection.

I still remember the day he lost his temper when one of the injector seatings was misaligned. He was frustrated, but not because of failure—because he cared. That was Siddharth Mandal. Passionate. Fearless. Uncompromising in his pursuit of excellence.

Mandal was not just an engineer; he was a force of nature. He questioned everything, pushed boundaries, and turned obstacles into stepping stones. His legacy isn’t just in the cars he helped build—it’s in the mindset he left behind.

To every Ashwa team member reading this: Challenge convention. Be relentless. Innovate without fear. If you carry even a fraction of Mandal’s passion, you won’t just build great cars—you’ll build great futures.

Now, go make history.”



Ankit Sahu

2010 BATCH



“Siddharth Mandal was a dear friend who was really focused and enthusiastic about the growth of Ashwa Racing. He shaped the project and continued supporting it till his last breath. His passion for excellence was commendable and his business acumen was truly inspiring. Rest in peace big Man!!”



Prashanth

2011 BATCH- ENGINE



“I first met Siddharth, or Mandal as we all called him, when I joined Ashwa in my first year and helped his team fabricate their car. Apart from all the jokes he cracked, I also had the privilege of learning a fair amount about the car and the many, many lessons learned by team Ashwa over the years. He taught me the workings and design of the car’s muffler during that time. Mandal was also instrumental in helping me build and deliver the business presentation at our competition and had a good way of teaching us by sharing examples from the past. It made sure the lessons stuck with us at that point.”



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ALUMNI CORNER

CONNECTING MEMORIES,
BUILDING FUTURES!



ALUMNI CORNER



Hi, I'm Darren. I was the chief engineer for the 2024 batch. From a very young age, I've always been into cars and racing, so joining Ashwa Racing was a no-brainer for me. The opportunity to build a race car from the ground up in college taught me invaluable lessons—lessons that I carry with me today, shaping me into a better engineer, a better teammate, and a better person.

From making critical decisions that helped shape my team's journey to those long nights in the garage, pushing to extract every bit of performance and reliability from the car, the experience molded me into someone very different from who I was when I first started college. It taught me the value of hard work, persistence, and thinking ahead.

All those moments with the team—building the car, driving it, fixing it—will always be something I cherish. And now, looking back, knowing how our story unfolded, I can confidently say I would do it all over again. No doubt.

Nevertheless, the "Ashwa Racing Experience" continues to inspire and produce incredible individuals year after year, ready to take on the industry. I'm happy to see the torch being passed on and the club progressing so well. Wishing you all the best guys!



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JUNIOR TRAINING SESSIONS

NURTURING TALENT,
SHAPING THE FUTURE.



JUNIOR TRAINING SEASON BEGINS AT ASHWA

The journey of innovation and engineering excellence continues at Ashwa as we welcome a new batch of aspiring engineers. The junior training season officially kicked off with a general briefing session on March 21st, led by Venkat Shreyas. This session provided the juniors with an insightful introduction to Ashwa's operations, its subsystems, and the exciting learning opportunities ahead.

The briefing covered the structured process of building a competition-ready vehicle, detailing the six crucial phases—pre-design, design, procurement, manufacturing, assembly, and testing—each spanning three months, making it an 18-month cycle. The juniors were also introduced to the team hierarchy, starting from the core leadership, which includes the Team Lead, Project Manager, and Chief Engineer, followed by the subsystem divisions.

With this foundation set, the intensive training sessions are set to begin on March 24th and will continue until April 29th. Over the next month, juniors will immerse themselves in hands-on learning, gaining technical expertise and practical insights essential for contributing to Ashwa's legacy of excellence.

As we embark on this journey, we look forward to shaping the next generation of engineers who will drive Ashwa forward. Let the learning begin!





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TECHNICAL INSIGHT

NURTURING TALENT,
SHAPING THE FUTURE.



SOLVING PROBLEMS AT THEIR SOURCE: ROOT CAUSE ANALYSIS EXPLAINED

Technical component failures are often multilayered and complex, with several contributing factors that might have led to their occurrence. Root Cause Analysis (RCA) techniques are employed by engineering teams in order to effectively remedy component failures. RCA is a vital skillset for every engineer as it provides a robust framework to eliminate design and manufacturing errors.

There are several approaches to RCA, some widely used methods are listed below:

- 5 Whys Method: This is suitable for simple problems when a clear understanding of the cause and effect is available. The principle states that any problem can be effectively analysed and addressed if 5 or more layers of causes are identified and evaluated

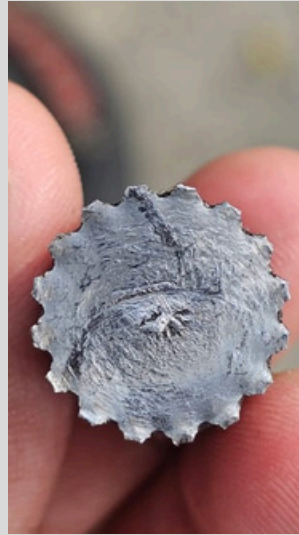
- Fishbone Diagram: When an issue may be caused by a variety of factors and errors, a successful approach is to analyse each potential cause through a fishbone diagram. The diagram enables succinct representation of all potential causes of a problem which allows for rigorous evaluation of all causes.

- Change Analysis: It is often useful to record and compare changes in the system before, during and after failure, which allows engineers to track any significant change that might have led to the failure. This approach provides a clear direction to the investigation and contributes to rapid convergence on the root cause and potential solutions.

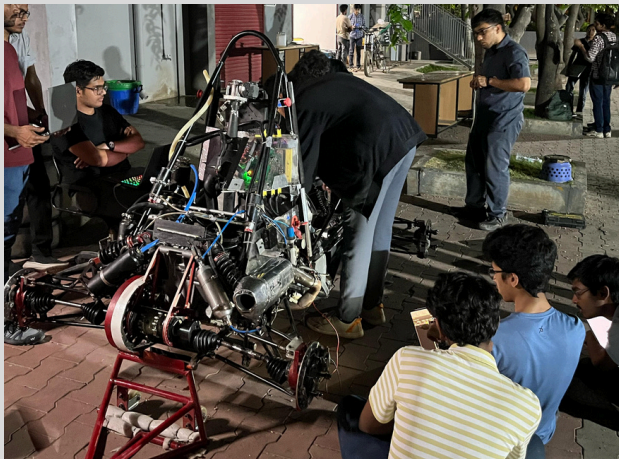
At Ashwa Mobility, our drivetrain team was tasked with determining the cause of recurring half shaft failures in RZ-XX5C, our combustion prototype which participated in Formula Bharat 2025.

The team used RCA principles and identified potential issues in the material quality, heat treating methods, as well as design flaws. Each potential cause was evaluated through material testing, simulations and calculations based on first principles. It was determined that increased design loads due to changes in the prototype's design as well as inadequate heat treatment most likely caused the failure of the shaft. Replacement shafts have been manufactured to resolve the issue.

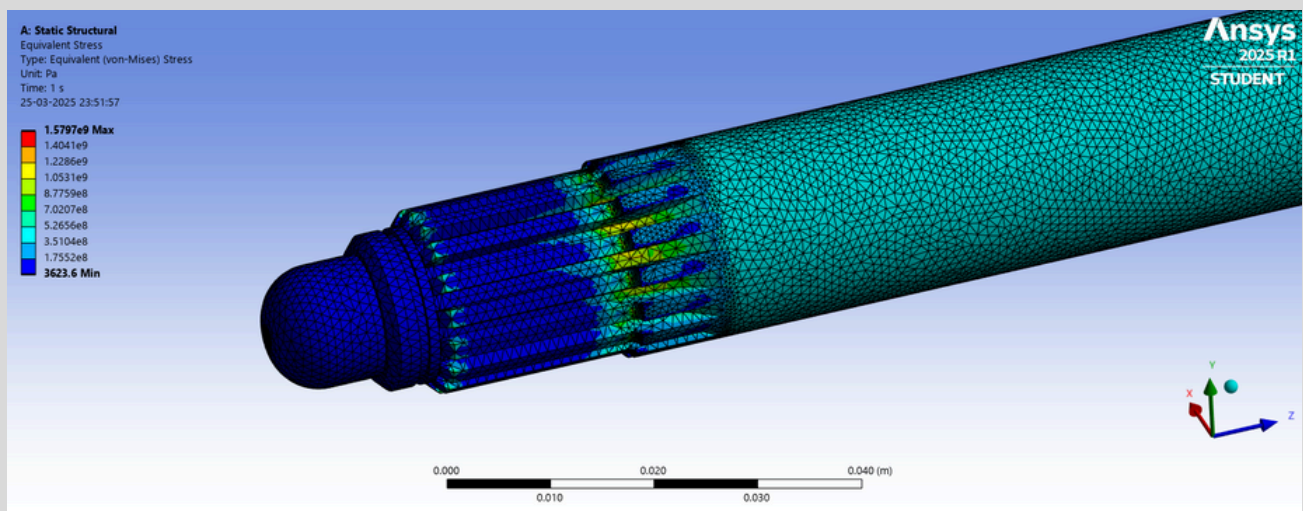
Although component failure is not ideal, we appreciate the opportunity to deepen our understanding of materials and processes while enhancing our skillset through Root Cause Analysis techniques.



We analyze the fracture point to assess its characteristics and determine the precise mode of failure, helping us understand the underlying causes and contributing factors.



After that, the loads acting on the shaft were calculated to gain a better understanding of the forces influencing its performance and contributing to the failure.



Finite Element Method (FEM) was applied to analyze the stress distribution on the shaft, allowing potential weak points to be identified and the root cause of the failure to be determined.



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Ansys



AKSHAYA MOTORS





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**TEAM
DIRECTORY
2025**

DRIVEN BY PASSION, UNITED BY ENGINEERING—
THIS IS ASHWA RACING!

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Sumeeth Anil Jadhav	Chassis and Workshop Management
Ipparla Ashakiran	Brakes and Logistics
Prathamesh Mali	IT
Pritam Panigrahi	Electrical & Testing
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